

Qualification Pack



Production Technician: Medical Device- Stents

Laser Cutting / Descaling and Annealing / Electropolishing/ Ultrasonic Cleaning
/ Drug Coating / Crimping/ Primary Packaging

QP Code: LFS/Q6202

Version: 1.0

NSQF Level: 5

Life Sciences Sector Skill Development Council || # 14, Rear 2nd Floor, Palam Marg, Vasant Vihar



Qualification Pack

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LFS/Q6202: Production Technician: Medical Device-Stents

Brief Job Description

The Production Technician for stent manufacturing is responsible for performing various processes like laser cutting, descaling, annealing, electropolishing, Ultrasonic Cleaning, drug coating and primary packaging of stent following Good Manufacturing Practices for the manufacturing of stent. The jobholder verifies the output to ensure that in batch manufacturing the quality parameters are met and ensures basic in-process quality checks for quality management. He/She also generates the critical records for every activity performed in compliance with data integrity rules. He/she also analyzes and report production/maintenance/quality related issues to the senior for solving work related problems. He/she also inspects products to verify conformance to specification.

Personal Attributes

The individual should have good communication skills in the regional language and able to comprehend the instructions and process documents in the English language. He/she should have good analytical skills. The job holder should be able to give attention to detail and understand the criticality of work.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [LFS/N6203: Perform pre-production check and preparation of various machines for manufacturing](#)
2. [LFS/N0130: Ensure adherence to environment, health and safety guidelines in manufacturing facility and controlled cleanroom area](#)
3. [LFS/N0131: Ensure a hygienic and clean work area to avoid contamination in manufacturing facility](#)
4. [LFS/N6211: Perform reporting and documentation with data integrity for medical devices and IVD](#)
5. [LFS/N0133: Coordinate and communicate with supervisor/ engineer, teams and auditors](#)
6. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Electives(mandatory to select at least one):

Elective 1: Laser Cutting

1. [LFS/N6204: Perform Laser Cutting Process](#)

Elective 2: Descaling and Annealing



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1. [LFS/N6205: Perform Descaling and Annealing process](#)

Elective 3: Electropolishing

1. [LFS/N6206: Perform Electropolishing Process](#)

Elective 4: Ultrasonic Cleaning

1. [LFS/N6207: Perform Ultrasonic Cleaning Process](#)

Elective 5: Drug Coating

1. [LFS/N6208: Perform Drug Coating Process](#)

Elective 6: Crimping

1. [LFS/N6209: Perform Crimping process](#)

Elective 7: Primary Packaging

1. [LFS/N6210: Perform Primary Packaging Process](#)

Qualification Pack (QP) Parameters

Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
Country	India
NSQF Level	5
Credits	38
Aligned to NCO/ISCO/ISIC Code	NCO-2015- 3139.1400



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Minimum Educational Qualification & Experience	Pursuing 2nd year of 2-year diploma after 12th (approved by AICTE or State Technical Education Regulatory Body in relevant field and continuous education) OR Pursuing 2nd year of UG (in life sciences or relevant fields)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	01/11/2026
NSQC Approval Date	01/11/2023
Version	1.0
Reference code on NQR	QG-5-LS-01272-2023-V1-LSSSDC
NQR Version	1

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LFS/N6203: Perform pre-production check and preparation of various machines for manufacturing

Description

This NOS is about performing pre-production checks and preparing machines for manufacturing operations

Scope

The scope covers the following :

- The NOS Scope are :
- Material Check
- Pre-production Safety Check
- Preparation of various machines for manufacturing

Elements and Performance Criteria

Material Check

To be competent, the user/individual on the job must be able to:

- PC1.** ensure stocks of required raw materials are ready and available as per minimum stock required
- PC2.** perform thorough inspection and verification of quality of all raw material
- PC3.** ensure that critical starting material, raw material are from a respective batch and analyzed by QC and approved by QA
- PC4.** match the Batch Code/ Item Code and Authorized Return (AR) No. of each compound/material with the Batch Code on the job schedule given by the planning department ensuring FIFO (First in First Out)
- PC5.** measure/weigh the starting material/ raw material/compound as per the desired specifications (shape, size and weight)
- PC6.** reach out to the supervisor for rejection control if disparities exist during visual inspection
- PC7.** return the unused material to warehouse with the appropriate label

Pre-production Safety check

To be competent, the user/individual on the job must be able to:

- PC8.** follow safe working instructions as per SOP and regulatory guidelines
- PC9.** ensure manufacturing process follows relevant safety regulations
- PC10.** perform comprehensive risk assessments to identify potential risks
- PC11.** Perform machinery and equipment basic function testing to ensure their proper working
- PC12.** Perform Job Safety Analysis in accordance with International/ National Standards
- PC13.** take secure permit/ authorization to work from appropriate authority

Preparation of various machines for manufacturing

To be competent, the user/individual on the job must be able to:

- PC14.** prepare machinery setup and configuration for stent manufacturing
- PC15.** ensure all the machines and equipments are calibrated & validated

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- PC16.** ensure that facilities, equipments and machinery are qualified by QA post cleaning validation and equipment validation
- PC17.** ensure proper loading of material into machine
- PC18.** Perform trial run and random tests to ensure accuracy
- PC19.** Perform routine maintenance of machinery and equipment
- PC20.** maintain history records and breakdown maintenance records
- PC21.** maintain electronic documentation for the trial runs and random test along with justifications for any wrong entries as applicable

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational SOPs relevant to pre-production safety check and preparation of machines for manufacturing operations
- KU2.** Purpose of in-process checks and their procedures
- KU3.** SOP for change request and maintenance of machine
- KU4.** procedures for documentation, reporting and escalation
- KU5.** relevant Data Integrity Rule and defined ISO Class guidelines
- KU6.** equipments used for broad level checks and their operative procedures
- KU7.** common issues in machine operations and their solutions
- KU8.** procedure for changeover cleaning
- KU9.** machinery and equipment calibration process
- KU10.** process for generating electronic records
- KU11.** methods for randoms tests in stent manufacturing
- KU12.** critical parameters for specific production machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record and communicate details of work done to appropriate people using written report or computer based record/electronic mail
- GS2.** understand the various coding systems as per company norms
- GS3.** communicate with cross functional team and supervisor
- GS4.** disclose information only to those who have the right and need to know it
- GS5.** suggest improvements(if any) in process based on experience
- GS6.** make decisions related to corrective maintenance and troubleshooting
- GS7.** analyse the readings of various operational variables and make informed decisions as per SOP
- GS8.** analyse any situation which needs an immediate escalation or emergency alarm

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Material Check</i>	10	15	5	2
PC1. ensure stocks of required raw materials are ready and available as per minimum stock required	-	-	-	-
PC2. perform thorough inspection and verification of quality of all raw material	-	-	-	-
PC3. ensure that critical starting material, raw material are from a respective batch and analyzed by QC and approved by QA	-	-	-	-
PC4. match the Batch Code/ Item Code and Authorized Return (AR) No. of each compound/material with the Batch Code on the job schedule given by the planning department ensuring FIFO (First in First Out)	-	-	-	-
PC5. measure/weigh the starting material/ raw material/compound as per the desired specifications (shape, size and weight)	-	-	-	-
PC6. reach out to the supervisor for rejection control if disparities exist during visual inspection	-	-	-	-
PC7. return the unused material to warehouse with the appropriate label	-	-	-	-
<i>Pre-production Safety check</i>	10	15	5	3
PC8. follow safe working instructions as per SOP and regulatory guidelines	-	-	-	-
PC9. ensure manufacturing process follows relevant safety regulations	-	-	-	-
PC10. perform comprehensive risk assessments to identify potential risks	-	-	-	-
PC11. Perform machinery and equipment basic function testing to ensure their proper working	-	-	-	-
PC12. Perform Job Safety Analysis in accordance with International/ National Standards	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take secure permit/ authorization to work from appropriate authority	-	-	-	-
<i>Preparation of various machines for manufacturing</i>	10	15	5	5
PC14. prepare machinery setup and configuration for stent manufacturing	-	-	-	-
PC15. ensure all the machines and equipments are calibrated & validated	-	-	-	-
PC16. ensure that facilities, equipments and machinery are qualified by QA post cleaning validation and equipment validation	-	-	-	-
PC17. ensure proper loading of material into machine	-	-	-	-
PC18. Perform trial run and random tests to ensure accuracy	-	-	-	-
PC19. Perform routine maintenance of machinery and equipment	-	-	-	-
PC20. maintain history records and breakdown maintenance records	-	-	-	-
PC21. maintain electronic documentation for the trial runs and random test along with justifications for any wrong entries as applicable	-	-	-	-
NOS Total	30	45	15	10



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6203
NOS Name	Perform pre-production check and preparation of various machines for manufacturing
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	2.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

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LFS/N0130: Ensure adherence to environment, health and safety guidelines in manufacturing facility and controlled cleanroom area

Description

This NOS is about ensuring adherence to environment, health and safety guidelines in manufacturing facility and controlled cleanroom area

Scope

The scope covers the following :

- The NOS scope are:
- Follow health and hygiene protocols
- Follow safety and security procedures
- Follow emergency procedures

Elements and Performance Criteria

Follow health and hygiene protocols

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and personal hygiene-related protocols as per regulatory guidelines/defined ISO class guidelines
- PC2.** ensure cleanroom facilities adhere to strict cleanliness guidelines
- PC3.** wash hands before entering in the production area as per SOP
- PC4.** report any allergy, sickness, or any other environment-related breach before or after entering the work premises to the designated person
- PC5.** ensure wearing cleanroom specialized garments while entering an environment-controlled work area
- PC6.** ensure proper disposal of used accessories

Follow safety and security procedures

To be competent, the user/individual on the job must be able to:

- PC7.** comply with safety and security policies and procedures
- PC8.** use appropriate safety gears like gowns, shoecover, headgear, masks, gloves and other relevant safety accessories as per regulatory guidelines throughout the process
- PC9.** use pulleys/ forklift (if available) to transport the material/ spares/ heavy assembly and tools
- PC10.** report any identified breaches in safety and security policies and procedures to the designated person
- PC11.** segregate material and follow the 5S system at the storage area
- PC12.** adhere to storage and handling guidelines for hazardous material and volatile chemicals
- PC13.** identify and correct any hazards that one can deal with safely, competently and within the limits of authority
- PC14.** ensure proper disposal of biohazardous material and chemical waste

Follow emergency procedures

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To be competent, the user/individual on the job must be able to:

- PC15.** raise the alarm and report any hazards which is not competent to deal with, to the relevant person in line with organizational procedures and warn other people who may get affected
- PC16.** inform the concerned person immediately about every unsafe act/ incident
- PC17.** follow emergency protocols efficiently

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** individual's role and responsibilities in maintaining healthy, hygienic, safe and secure working environment
- KU2.** company's procedures and protocols for the environment, health and safety
- KU3.** relevant legislative requirements as per local laws
- KU4.** implications that any non-compliance with health, safety and security may have on individuals and the organization
- KU5.** workplace hazards in the manufacturing/ research facility in the life sciences sector and reporting procedures for unhealthy/ unsafe act/incidents, hazards and accident as per defined ISO class
- KU6.** limits of individual responsibility for dealing with hazards
- KU7.** chemical substances, their characteristics, and required precaution and safety measures
- KU8.** cleanroom specialized garment
- KU9.** the organization's emergency procedures for different emergency situations and the importance of following these procedures
- KU10.** evacuation procedures for employees, contract staff and visitors
- KU11.** procedure to summon medical assistance and the emergency services, where necessary
- KU12.** different types of breaches in the environment, health, safety and security and how and when to report these
- KU13.** Indian Medical device Rules 2017 and ISO 13485 guidelines for personal hygiene
- KU14.** types of safety gears and procedure to use them
- KU15.** importance of material segregation and 5S system
- KU16.**
 - importance of material segregation and 5S system
 - guidelines for handling and storing hazardous material

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to read instructions, guidelines, procedures, rules, and signages
- GS2.** use written communication skills to accurately record every information required to be reported as per SOP and defined ISO class guidelines in the language prescribed by the company's SOP



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- GS3.** apply planning and organizing skills to plan and organize tools and material required for work to fulfil environment, health, safety and security requirements
- GS4.** use critical thinking skills to ascertain the breach/ compliance of EHS protocols
- GS5.** apply customer centricity to remain compliant with data integrity rules, defined ISO class guidelines and to evaluate impact of wrongdoings
- GS6.** apply decision-making skills to make balanced judgments within the authority in different situations while dealing with hazards and breaches

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and hygiene protocols</i>	15	15	5	5
PC1. comply with health and personal hygiene-related protocols as per regulatory guidelines/defined ISO class guidelines	-	-	-	-
PC2. ensure cleanroom facilities adhere to strict cleanliness guidelines	-	-	-	-
PC3. wash hands before entering in the production area as per SOP	-	-	-	-
PC4. report any allergy, sickness, or any other environment-related breach before or after entering the work premises to the designated person	-	-	-	-
PC5. ensure wearing cleanroom specialized garments while entering an environment-controlled work area	-	-	-	-
PC6. ensure proper disposal of used accessories	-	-	-	-
<i>Follow safety and security procedures</i>	15	15	3	3
PC7. comply with safety and security policies and procedures	-	-	-	-
PC8. use appropriate safety gears like gowns, shoe cover, headgear, masks, gloves and other relevant safety accessories as per regulatory guidelines throughout the process	-	-	-	-
PC9. use pulleys/ forklift (if available) to transport the material/ spares/ heavy assembly and tools	-	-	-	-
PC10. report any identified breaches in safety and security policies and procedures to the designated person	-	-	-	-
PC11. segregate material and follow the 5S system at the storage area	-	-	-	-
PC12. adhere to storage and handling guidelines for hazardous material and volatile chemicals	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. identify and correct any hazards that one can deal with safely, competently and within the limits of authority	-	-	-	-
PC14. ensure proper disposal of biohazardous material and chemical waste	-	-	-	-
<i>Follow emergency procedures</i>	10	10	2	2
PC15. raise the alarm and report any hazards which is not competent to deal with, to the relevant person in line with organizational procedures and warn other people who may get affected	-	-	-	-
PC16. inform the concerned person immediately about every unsafe act/ incident	-	-	-	-
PC17. follow emergency protocols efficiently	-	-	-	-
NOS Total	40	40	10	10



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0130
NOS Name	Ensure adherence to environment, health and safety guidelines in manufacturing facility and controlled cleanroom area
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Generic
NSQF Level	4.5
Credits	1.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023



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LFS/N0131: Ensure a hygienic and clean work area to avoid contamination in manufacturing facility

Description

This NOS is about ensuring a hygienic and clean work area to avoid contamination

Scope

The scope covers the following :

- The NOS scope are:
- Sanitation activities before the start of work
- Sanitation activities during work
- Sanitation activities after completion of work

Elements and Performance Criteria

Sanitation activities before the start of work

To be competent, the user/individual on the job must be able to:

- PC1.** ensure sanitation of cleanroom area as per SOP
- PC2.** check for validation tag on machines and equipments
- PC3.** ensure proper cleaning of machinery and equipments
- PC4.** ensure cleaning validation in the presence of authorized personnel or QA inspector
- PC5.** ensure adequate ventilation during work being carried out
- PC6.** ensure regular sterilization of machinery and equipment
- PC7.** handle the cleaning material/reagent only after wearing the personal protective equipment required for the cleaning method
- PC8.** segregate and store the material with an appropriate label in designated places to avoid contamination

Sanitation activities during work

To be competent, the user/individual on the job must be able to:

- PC9.** segregate and store the intermediate material with an appropriate label in designated places to avoid contamination
- PC10.** report any additional cleaning requirement that is outside one's purview, to the appropriate person
- PC11.** segregate, store and dispose of the rejected products or generated waste as per SOP under the supervision of supervisor and EHS personnel

Sanitation activities after completion of work

To be competent, the user/individual on the job must be able to:

- PC12.** ensure proper disposal of hazardous material and chemical waste
- PC13.** ensure removal of scrap material from work area
- PC14.** ensure cleaning of the equipment after every batch production as per SOP



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- PC15.** perform the cleaning validation of the equipment in the presence of designated authorized personnel and QA inspector
- PC16.** ensure clean and dry area after completion of work as per ISO requirement of sanitized premises

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** levels of hygiene required by production area and the importance of maintaining the same
- KU2.** methodology for production area inspection with methods and materials required for cleaning a variety of surfaces and equipment
- KU3.** methodology to check machinery and equipment on completion of cleaning
- KU4.** procedures for reporting any unidentified soiling or any deviation for cleaning validation
- KU5.** role of different materials, machinery and equipment in cleaning and sanitation of manufacturing area
- KU6.** defined ISO class guidelines for cleaning/ sanitation activity and maintaining hygiene
- KU7.** cleaning validation process

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use written communication skills to record and communicate details of work done to appropriate people using written/typed report and electronic mail
- GS2.** use verbal communication skills to communicate with supervisor, cross-functional teams and auditors effectively
- GS3.** use critical thinking skills to interpret the various coding systems as per company norms and in identifying the non-compliance while performing the area inspection
- GS4.** apply customer centricity at work
- GS5.** apply problem-solving and decision making while dealing with any deviation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sanitation activities before the start of work</i>	15	15	5	5
PC1. ensure sanitation of cleanroom area as per SOP	-	-	-	-
PC2. check for validation tag on machines and equipments	-	-	-	-
PC3. ensure proper cleaning of machinery and equipments	-	-	-	-
PC4. ensure cleaning validation in the presence of authorized personnel or QA inspector	-	-	-	-
PC5. ensure adequate ventilation during work being carried out	-	-	-	-
PC6. ensure regular sterilization of machinery and equipment	-	-	-	-
PC7. handle the cleaning material/reagent only after wearing the personal protective equipment required for the cleaning method	-	-	-	-
PC8. segregate and store the material with an appropriate label in designated places to avoid contamination	-	-	-	-
<i>Sanitation activities during work</i>	15	15	3	3
PC9. segregate and store the intermediate material with an appropriate label in designated places to avoid contamination	-	-	-	-
PC10. report any additional cleaning requirement that is outside one's purview, to the appropriate person	-	-	-	-
PC11. segregate, store and dispose of the rejected products or generated waste as per SOP under the supervision of supervisor and EHS personnel	-	-	-	-
<i>Sanitation activities after completion of work</i>	10	10	2	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure proper disposal of hazardous material and chemical waste	-	-	-	-
PC13. ensure removal of scrap material from work area	-	-	-	-
PC14. ensure cleaning of the equipment after every batch production as per SOP	-	-	-	-
PC15. perform the cleaning validation of the equipment in the presence of designated authorized personnel and QA inspector	-	-	-	-
PC16. ensure clean and dry area after completion of work as per ISO requirement of sanitized premises	-	-	-	-
NOS Total	40	40	10	10



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0131
NOS Name	Ensure a hygienic and clean work area to avoid contamination in manufacturing facility
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Generic
NSQF Level	4.5
Credits	1.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023



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LFS/N6211: Perform reporting and documentation with data integrity for medical devices and IVD

Description

This NOS is about performing reporting and documentation with data integrity for medical devices and IVD

Scope

The scope covers the following :

- The NOS scope are :
- Reporting and escalation of deviations
- Documentation

Elements and Performance Criteria

Reporting and escalation of deviations

To be competent, the user/individual on the job must be able to:

- PC1.** identify non-conformities to quality assurance standards and product specifications
- PC2.** identify potential causes of non-conformities to quality assurance standards with the help of supervisor
- PC3.** implement the corrective and preventive actions as guided by the engineer and quality assurance team

Documentation

To be competent, the user/individual on the job must be able to:

- PC4.** identify documentation to be completed as per SOP and ISO class rules
- PC5.** record the required information of all significant activities, incidents and deviations as per recording formats in compliance with SOP and defined ISO class guidelines
- PC6.** maintain both electronic and manual records in the log books and other documentation required as per defined ISO class
- PC7.** update the Equipment log books , status boards & ensure in line with the process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** procedures for documentation, reporting, and escalation of incidents and deviations
- KU2.** procedure for generating manual and electronic records
- KU3.** ALCOA Principles, Data integrity, and information security rules
- KU4.** procedure for line clearance

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use reading and comprehension skills to read the various coding systems, instructions, guidelines, procedures, rules, and signages
- GS3.** use verbal communication skills to interact with teammates, and during interaction with regulatory inspectors and other people
- GS4.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS5.** apply critical thinking skills to analyze and identify what and when to report an issue to the concerned supervisor and to deal with a colleague individually, depending on the type of concern
- GS6.** use critical thinking skills in analyzing the impact of deviations, wastage, and rejects to the safety, environment and efficiency, compliance, and cost
- GS7.** apply customer-centricity to remain compliant with data integrity rules, and while responding to auditors and QA personnel

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Reporting and escalation of deviations</i>	20	20	5	5
PC1. identify non-conformities to quality assurance standards and product specifications	-	-	-	-
PC2. identify potential causes of non-conformities to quality assurance standards with the help of supervisor	-	-	-	-
PC3. implement the corrective and preventive actions as guided by the engineer and quality assurance team	-	-	-	-
<i>Documentation</i>	20	20	5	5
PC4. identify documentation to be completed as per SOP and ISO class rules	-	-	-	-
PC5. record the required information of all significant activities, incidents and deviations as per recording formats in compliance with SOP and defined ISO class guidelines	-	-	-	-
PC6. maintain both electronic and manual records in the log books and other documentation required as per defined ISO class	-	-	-	-
PC7. update the Equipment log books , status boards & ensure in line with the process	-	-	-	-
NOS Total	40	40	10	10



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6211
NOS Name	Perform reporting and documentation with data integrity for medical devices and IVD
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023



Qualification Pack

LFS/N0133: Coordinate and communicate with supervisor/ engineer, teams and auditors

Description

This NOS unit is about coordinating and communicating with supervisor/ engineer, teams and auditors

Scope

The scope covers the following :

- Coordination with supervisor/engineer
- Team coordination
- Respond to audit queries
- Sensitivity towards all genders and people with disability

Elements and Performance Criteria

Coordination with supervisor/engineer

To be competent, the user/individual on the job must be able to:

- PC1.** follow work instructions received from the reporting supervisor
- PC2.** adhere to the production plan to meet timelines
- PC3.** communicate process-flow improvements, production defects received from the previous process, repairs and maintenance of equipment to reporting supervisor
- PC4.** ensure timely intimation to supervisor/ production engineer about planned absence/ illness/ dizziness during work/ critical issues requiring his/her intervention
- PC5.** engage with supervisor on work-related and behavioral feedback and demonstrate continual learning and self-direction

Team coordination

To be competent, the user/individual on the job must be able to:

- PC6.** support team members and colleagues of other departments in work
- PC7.** take handover from previous shift operator and give handover to next shift operator as per SOP
- PC8.** guide manufacturing and packaging assistants during manufacturing process
- PC9.** coordinate with warehouse team for material dispensing and issuance
- PC10.** find solutions for workflow related difficulties with mutual agreement
- PC11.** coordinate with maintenance team for preventive and corrective maintenance, break down and calibration errors
- PC12.** coordinate with quality control team for sample collection, batch release and in-process checks
- PC13.** coordinate with quality assurance team for machine/ equipment validation at a routine interval as per SOP
- PC14.** provide inputs to the concerned stakeholders in periodic fence line review to detect non-compliance

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PC15. coordinate with EHS team for any safety incident, accident and emergency

Respond to audit queries

To be competent, the user/individual on the job must be able to:

PC16. ensure interact with auditor in a preferable language

PC17. provide an honest and clear answer to the raised queries

PC18. maintain data integrity while responding to auditors or while producing the documented records for activities performed

Sensitivity towards all genders and people with disability

To be competent, the user/individual on the job must be able to:

PC19. respect all genders, religions, and caste

PC20. empathize with the people with disability

PC21. offer support or help to a person with disability only when asked

PC22. ensure to adhere with the guidelines laid in Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act

PC23. report any violation of prevention of sexual harassment (POSH) rules immediately to the POSH committee

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the company's policies on the preferred language of communication, reporting and escalation policy

KU2. the reporting structure of the organization

KU3. types of audits in the life sciences sector for the manufacturing operations

KU4. the required regulatory and statutory compliance-related documentation

KU5. the guidelines for data integrity, ethics, and compliance in the life sciences industry

KU6. the guidelines laid on Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act

KU7. the methods of workplace communication

KU8. importance of team coordination

KU9. the types of possible disabilities among people with disability (PwD)

KU10. the challenges faced by PwD

KU11. the importance of displaying empathy towards PwD

KU12. the right way to use the laws, acts, and provisions defined for PwD by the statutory bodies

KU13. the importance of awareness for gender sensitization and prevention of sexual harassment (POSH) act

KU14. importance of respecting all gender identities, religion, caste, and culture

KU15. method to receive the performance feedback

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** use reading and comprehension skills to gauge the relevant information manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/ comments
- GS2.** use written communication skills to record and communicate details of work done to appropriate stakeholders by using written or computer-based record/ electronic mail in a given format and compliant with ALCOA principle
- GS3.** use verbal communication skills to communicate confidential and sensitive information discretely to the authorized person while interacting with teammates
- GS4.** use team-building skills during the interaction with teammates while managing the difficult/stressful or emotional situations at work
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical thinking skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** apply customer-centricity skills while responding to auditors and QA personnel

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Coordination with supervisor/engineer</i>	10	10	4	4
PC1. follow work instructions received from the reporting supervisor	-	-	-	-
PC2. adhere to the production plan to meet timelines	-	-	-	-
PC3. communicate process-flow improvements, production defects received from the previous process, repairs and maintenance of equipment to reporting supervisor	-	-	-	-
PC4. ensure timely intimation to supervisor/ production engineer about planned absence/ illness/ dizziness during work/ critical issues requiring his/her intervention	-	-	-	-
PC5. engage with supervisor on work-related and behavioral feedback and demonstrate continual learning and self-direction	-	-	-	-
<i>Team coordination</i>	10	10	2	2
PC6. support team members and colleagues of other departments in work	-	-	-	-
PC7. take handover from previous shift operator and give handover to next shift operator as per SOP	-	-	-	-
PC8. guide manufacturing and packaging assistants during manufacturing process	-	-	-	-
PC9. coordinate with warehouse team for material dispensing and issuance	-	-	-	-
PC10. find solutions for workflow related difficulties with mutual agreement	-	-	-	-
PC11. coordinate with maintenance team for preventive and corrective maintenance, break down and calibration errors	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. coordinate with quality control team for sample collection, batch release and in-process checks	-	-	-	-
PC13. coordinate with quality assurance team for machine/ equipment validation at a routine interval as per SOP	-	-	-	-
PC14. provide inputs to the concerned stakeholders in periodic fence line review to detect non-compliance	-	-	-	-
PC15. coordinate with EHS team for any safety incident, accident and emergency	-	-	-	-
<i>Respond to audit queries</i>	10	10	2	2
PC16. ensure interact with auditor in a preferable language	-	-	-	-
PC17. provide an honest and clear answer to the raised queries	-	-	-	-
PC18. maintain data integrity while responding to auditors or while producing the documented records for activities performed	-	-	-	-
<i>Sensitivity towards all genders and people with disability</i>	10	10	2	2
PC19. respect all genders, religions, and caste	-	-	-	-
PC20. empathize with the people with disability	-	-	-	-
PC21. offer support or help to a person with disability only when asked	-	-	-	-
PC22. ensure to adhere with the guidelines laid in Sexual Harassment of Women at Workplace(Prevention, Prohibition and Redressal) Act	-	-	-	-
PC23. report any violation of prevention of sexual harassment (POSH) rules immediately to the POSH committee	-	-	-	-
NOS Total	40	40	10	10



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National Occupational Standards (NOS) Parameters

NOS Code	LFS/N0133
NOS Name	Coordinate and communicate with supervisor/ engineer, teams and auditors
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023



Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

LFS/N6204: Perform Laser Cutting Process

Description

The NOS is about performing laser cutting process

Scope

The scope covers the following :

- The NOS Scope is :
- Laser Cutting Operations

Elements and Performance Criteria

Laser Cutting Operations

To be competent, the user/individual on the job must be able to:

- PC1.** ensure availability of high quality raw material as per defined ISO Class
- PC2.** charge the identified material in the correct pattern as per the Batch Record to minimize material overflow/ wastage/ excess flash
- PC3.** ensure set up of laser cutting system including high power laser source, beam delivery system and computer controlled CNC (Computer Numerical Control)
- PC4.** perform loading of cobalt chromium tube properly into laser machine
- PC5.** perform securing of material within CNC machine
- PC6.** perform laser cutting for creation of intricate stent design with high accuracy
- PC7.** ensure set up of cooling or gas system to prevent excessive heat buildup and oxidation of the metal
- PC8.** perform troubleshooting of machinery and equipment
- PC9.** minimize waste/ rejections during entire process
- PC10.** ensure availability of high quality raw material as per defined ISO Class

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** clean room behaviour practices and gowning procedures
- KU2.** SOP for entry and exit from manufacturing area
- KU3.** PPE used in laser cutting process and their work instructions
- KU4.** material segregation, handling and storage guidelines
- KU5.** type of machinery and equipment for laser cutting process and critical parameters
- KU6.** care and maintenance of equipment and machinery
- KU7.** removing and replacing components or parts of machinery and equipment
- KU8.** specialized techniques used in laser cutting process
- KU9.** utilities used in laser cutting process



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- KU10.** basic in-process checks for laser cutting process
- KU11.** raw material used in laser cutting process
- KU12.** procedure of set up of laser cutting system
- KU13.** procedure of loading material into machinery
- KU14.** Computer Numerical Control Machine
- KU15.** procedure of set up of cooling or gas system
- KU16.** procedure of troubleshooting equipments and manufacturing issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to maintain proper and concise records as per given format and compliant with ALCOA principle
- GS3.** use verbal communication skills in communicating the details of work done to appropriate people and during interaction with regulatory inspectors and other teammates
- GS4.** use planning and organizing skills in every activity planned and performed in stent operations and to achieve resource optimization
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** use critical skills in analyzing impact of deviations, wastage and rejects to the environment and efficiency, complianc and cost
- GS8.** demonstrate decision making while making necessary adjustments in parameters to achieve quality specifications
- GS9.** apply customer-centricity while responding to auditors and QA personnel

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Laser Cutting Operations</i>	30	50	10	10
PC1. ensure availability of high quality raw material as per defined ISO Class	-	-	-	-
PC2. charge the identified material in the correct pattern as per the Batch Record to minimize material overflow/ wastage/ excess flash	-	-	-	-
PC3. ensure set up of laser cutting system including high power laser source, beam delivery system and computer controlled CNC (Computer Numerical Control)	-	-	-	-
PC4. perform loading of cobalt chromium tube properly into laser machine	-	-	-	-
PC5. perform securing of material within CNC machine	-	-	-	-
PC6. perform laser cutting for creation of intricate stent design with high accuracy	-	-	-	-
PC7. ensure set up of cooling or gas system to prevent excessive heat buildup and oxidation of the metal	-	-	-	-
PC8. perform troubleshooting of machinery and equipment	-	-	-	-
PC9. minimize waste/ rejections during entire process	-	-	-	-
PC10. ensure availability of high quality raw material as per defined ISO Class	-	-	-	-
NOS Total	30	50	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6204
NOS Name	Perform Laser Cutting Process
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	4.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

LFS/N6205: Perform Descaling and Annealing process

Description

The NOS is about performing descaling and annealing process

Scope

The scope covers the following :

- The NOS Scope is :
- Descaling Operations
- Annealing Operations

Elements and Performance Criteria

Descaling Operations

To be competent, the user/individual on the job must be able to:

- PC1.** ensure set up of descaling machinery
- PC2.** perform descaling process to remove unwanted metal oxide, slag, burrs, debris etc.
- PC3.** perform immersing of laser cut stent into an acidic descaling solution
- PC4.** perform applying ultrasonic waves to solution containing stent
- PC5.** ensure visual inspection of Descaled stent
- PC6.** ensure passed descaled stent are placed in designated container for annealing process

Annealing Operations

To be competent, the user/individual on the job must be able to:

- PC7.** perform set up of annealing machinery
- PC8.** PC7. Perform annealing process to soften the metal and relieve internal stress in metal created during laser cutting process
- PC9.** perform loading of descaled stent into quartz crucible which is subjected to thermal cycle inside annealing furnace
- PC10.** perform controlling of thermal cycle through a computer software
- PC11.** perform taking out stent from annealing furnace and immerse in DI water which is subjected to ultrasonic waves
- PC12.** ensure visual inspection of annealed stents
- PC13.** ensure passed annealed stents are placed in designated container and transferred for electropolishing process
- PC14.** perform troubleshooting of machinery and equipment
- PC15.** minimize waste/ rejections during entire process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** clean room behaviour practices and gowning procedures
- KU2.** SOP for entry and exit from manufacturing area
- KU3.** PPE used in descaling and annealing process and their work instructions
- KU4.** material segregation, handling and storage guidelines
- KU5.** type of machinery and equipment for descaling and annealing process and critical parameters
- KU6.** care and maintenance of equipment and machinery
- KU7.** removing and replacing components or parts of machinery and equipment
- KU8.** specialized techniques used in descaling and annealing process
- KU9.** utilities used in descaling and annealing process
- KU10.** basic in-process checks for descaling and annealing process
- KU11.** raw material used in descaling and annealing process
- KU12.** basic knowledge of ALCOA Principles
- KU13.** procedure of set up of descaling and annealing machinery
- KU14.** procedure of descaling
- KU15.** procedure of annealing
- KU16.** procedure of thermal cycle
- KU17.** Quartz Crucible and Annealing Furnace
- KU18.** procedure of troubleshooting equipments and manufacturing issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to maintain proper and concise records as per given format and compliant with ALCOA principle
- GS3.** use verbal communication skills in communicating the details of work done to appropriate people and during interaction with regulatory inspectors and other teammates
- GS4.** use planning and organizing skills in every activity planned and performed in stent operations and to achieve resource optimization
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** use critical skills in analyzing impact of deviations, wastage and rejects to the environment and efficiency, compliance and cost
- GS8.** demonstrate decision making while making necessary adjustments in parameters to achieve quality specifications
- GS9.** apply customer-centricity while responding to auditors and QA personnel

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Descaling Operations</i>	15	25	5	5
PC1. ensure set up of descaling machinery	-	-	-	-
PC2. perform descaling process to remove unwanted metal oxide, slag, burrs, debris etc.	-	-	-	-
PC3. perform immersing of laser cut stent into an acidic descaling solution	-	-	-	-
PC4. perform applying ultrasonic waves to solution containing stent	-	-	-	-
PC5. ensure visual inspection of Descaled stent	-	-	-	-
PC6. ensure passed descaled stent are placed in designated container for annealing process	-	-	-	-
<i>Annealing Operations</i>	15	25	5	5
PC7. perform set up of annealing machinery	-	-	-	-
PC8. PC7. Perform annealing process to soften the metal and relieve internal stress in metal created during laser cutting process	-	-	-	-
PC9. perform loading of descaled stent into quartz crucible which is subjected to thermal cycle inside annealing furnace	-	-	-	-
PC10. perform controlling of thermal cycle through a computer software	-	-	-	-
PC11. perform taking out stent from annealing furnace and immerse in DI water which is subjected to ultrasonic waves	-	-	-	-
PC12. ensure visual inspection of annealed stents	-	-	-	-
PC13. ensure passed annealed stents are placed in designated container and transferred for electropolishing process	-	-	-	-
PC14. perform troubleshooting of machinery and equipment	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. minimize waste/ rejections during entire process	-	-	-	-
NOS Total	30	50	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6205
NOS Name	Perform Descaling and Annealing process
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	4.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023



Qualification Pack

LFS/N6206: Perform Electropolishing Process

Description

The NOS is about performing electropolishing process

Scope

The scope covers the following :

- The NOS Scope is :
- Electropolishing Operations

Elements and Performance Criteria

Electropolishing Operations

To be competent, the user/individual on the job must be able to:

- PC1.** perform electropolishing process to achieve ultra-smooth shiny surface and final stent dimension
- PC2.** ensure temperature of electrolyte is maintained as per SOP
- PC3.** ensure proper composition of metal alloy undergoing electropolishing as per SOP
- PC4.** ensure proper electrolyte composition as per SOP
- PC5.** perform loading of metal stent over a holder and immerse it into liquid electrolytic solution along with a metal cathode
- PC6.** perform applying electric current flow through stent
- PC7.** ensure density and time of exposure of electric current as per SOP
- PC8.** ensure preferential removal of protruding portions of surface structure to get smoother metal surface
- PC9.** perform visual inspection and maintain dimensional measurement of electropolished stent
- PC10.** ensure passed electropolished stents are placed in a designated container

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** clean room behaviour practices and gowning procedures
- KU2.** SOP for entry and exit from manufacturing area
- KU3.** PPE used in electropolishing process and their work instructions
- KU4.** material segregation, handling and storage guidelines
- KU5.** type of machinery and equipment used for electropolishing and their critical parameters
- KU6.** care and maintenance of equipment and machinery
- KU7.** removing and replacing components or parts of machinery and equipment
- KU8.** specialized techniques used in electropolishing process
- KU9.** utilities used in electropolishing process



Qualification Pack

- KU10.** basic in-process checks for electropolishing process
- KU11.** procedure of set up of electropolishing system
- KU12.** procedure of electropolishing
- KU13.** procedure of troubleshooting equipments and manufacturing issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to maintain proper and concise records as per given format and compliant with ALCOA principle
- GS3.** use verbal communication skills in communicating the details of work done to appropriate people and during interaction with regulatory inspectors and other teammates
- GS4.** use planning and organizing skills in every activity planned and performed in stent operations and to achieve resource optimization
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** use critical skills in analyzing impact of deviations, wastage and rejects to the environment and efficiency, compliance and cost
- GS8.** demonstrate decision making while making necessary adjustments in parameters to achieve quality specifications
- GS9.** apply customer-centricity while responding to auditors and QA personnel

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Electropolishing Operations</i>	30	50	10	10
PC1. perform electropolishing process to achieve ultra-smooth shiny surface and final stent dimension	-	-	-	-
PC2. ensure temperature of electrolyte is maintained as per SOP	-	-	-	-
PC3. ensure proper composition of metal alloy undergoing electropolishing as per SOP	-	-	-	-
PC4. ensure proper electrolyte composition as per SOP	-	-	-	-
PC5. perform loading of metal stent over a holder and immerse it into liquid electrolytic solution along with a metal cathode	-	-	-	-
PC6. perform applying electric current flow through stent	-	-	-	-
PC7. ensure density and time of exposure of electric current as per SOP	-	-	-	-
PC8. ensure preferential removal of protruding portions of surface structure to get smoother metal surface	-	-	-	-
PC9. perform visual inspection and maintain dimensional measurement of electropolished stent	-	-	-	-
PC10. ensure passed electropolished stents are placed in a designated container	-	-	-	-
NOS Total	30	50	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6206
NOS Name	Perform Electropolishing Process
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	4.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023



Qualification Pack

LFS/N6207: Perform Ultrasonic Cleaning Process

Description

The NOS is about performing ultrasonic cleaning process

Scope

The scope covers the following :

- The NOS Scope is :
- Ultrasonic Cleaning Operations

Elements and Performance Criteria

Ultrasonic Cleaning Operations

To be competent, the user/individual on the job must be able to:

- PC1.** ensure cleanroom and equipments are clean to avoid cross contamination
- PC2.** follow strict hygiene protocols including wearing appropriate protective gear
- PC3.** ensure cleaning solutions contains wetting agents to reduce surface tension and promote cavitation for proper cleaning
- PC4.** perform ultrasonic cleaning process to ensure safety and effectiveness of stent
- PC5.** ensure proper ultrasonic bath in ultrasonic cleaning tank
- PC6.** ensure Electronic Transducer generate equal and consistent distribution of sound waves in cleaning solution to facilitate cleaning action
- PC7.** ensure Electronic Generator provides electrical energy needed to generate high frequency mechanical vibrations to induce cavitation in cleaning solution
- PC8.** perform multiple rinse with purified water to remove any cleaning solution on stent
- PC9.** perform air drying or use of controlled oven to remove residual moisture and preventing from corrosion
- PC10.** ensure removal of defective stent from production line

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** clean room behaviour practices and gowning procedures
- KU2.** SOP for entry and exit from manufacturing area
- KU3.** PPE used in ultrasonic cleaning process and their work instructions
- KU4.** material segregation, handling and storage guidelines
- KU5.** type of machinery and equipment used for ultrasonic cleaning process and critical parameters
- KU6.** care and maintenance of equipment and machinery
- KU7.** removing and replacing components or parts of machinery and equipment



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- KU8.** specialized techniques used in ultrasonic cleaning of stent
- KU9.** utilities used in ultrasonic cleaning process
- KU10.** basic in-process checks for ultrasonic cleaning process

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to maintain proper and concise records as per given format and compliant with ALCOA principle
- GS3.** use verbal communication skills in communicating the details of work done to appropriate people and during interaction with regulatory inspectors and other teammates
- GS4.** use planning and organizing skills in every activity planned and performed in stent operations and to achieve resource optimization
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** use critical skills in analyzing impact of deviations, wastage and rejects to the environment and efficiency, compliance and cost
- GS8.** demonstrate decision making while making necessary adjustments in parameters to achieve quality specifications
- GS9.** apply customer-centricity while responding to auditors and QA personnel

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ultrasonic Cleaning Operations</i>	30	50	10	10
PC1. ensure cleanroom and equipments are clean to avoid cross contamination	-	-	-	-
PC2. follow strict hygiene protocols including wearing appropriate protective gear	-	-	-	-
PC3. ensure cleaning solutions contains wetting agents to reduce surface tension and promote cavitation for proper cleaning	-	-	-	-
PC4. perform ultrasonic cleaning process to ensure safety and effectiveness of stent	-	-	-	-
PC5. ensure proper ultrasonic bath in ultrasonic cleaning tank	-	-	-	-
PC6. ensure Electronic Transducer generate equal and consistent distribution of sound waves in cleaning solution to facilitate cleaning action	-	-	-	-
PC7. ensure Electronic Generator provides electrical energy needed to generate high frequency mechanical vibrations to induce cavitation in cleaning solution	-	-	-	-
PC8. perform multiple rinse with purified water to remove any cleaning solution on stent	-	-	-	-
PC9. perform air drying or use of controlled oven to remove residual moisture and preventing from corrosion	-	-	-	-
PC10. ensure removal of defective stent from production line	-	-	-	-
NOS Total	30	50	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6207
NOS Name	Perform Ultrasonic Cleaning Process
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	4.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

LFS/N6208: Perform Drug Coating Process

Description

The NOS is about performing drug coating process

Scope

The scope covers the following :

- The NOS Scope is :
- Preparation of Drug Coating Operations
- Drug Coating Operations

Elements and Performance Criteria

Preparation of Drug Coating Operations

To be competent, the user/individual on the job must be able to:

- PC1.** ensure availability of high quality raw material as per defined ISO Class
- PC2.** perform preparation of drug coating solution under reverse laminar air in dispensing booth of class 100 (ISO class 5) surrounded by class 10000 (ISO class 8)
- PC3.** ensure drug coating solution is prepared as per SOP
- PC4.** ensure cleanroom are periodically validated and periodic environmental monitoring is performed as per SOP
- PC5.** ensure drug coating solution is tested as per quality plan

Drug Coating Operations

To be competent, the user/individual on the job must be able to:

- PC6.** ensure coating process is performed with a validated process in a controlled environment of class 10000
- PC7.** ensure pre cleaned stent identification number and its measured weight are recorded accurately
- PC8.** perform pouring of prepared solution into spray gun and start coating process
- PC9.** ensure vacuum drying process is performed in same controlled environment of class 10000
- PC10.** ensure cleanroom are periodically validated
- PC11.** ensure drug coating process is validated as per records
- PC12.** ensure sending the drug coated stent to next stage after quality inspection as per quality plan
- PC13.** ensure any rejected drug coated stents are modified if feasible else segregate and dispose them as per SOP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** clean room behaviour practices and gowning procedures

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- KU2.** SOP for entry and exit from manufacturing area
- KU3.** PPE used in descaling and annealing process and their work instructions
- KU4.** material segregation, handling and storage guidelines
- KU5.** type of machinery and equipment for drug coating process of stent and critical parameters
- KU6.** care and maintenance of equipment and machinery
- KU7.** removing and replacing components or parts of machinery and equipment
- KU8.** specialized techniques used in drug coating process
- KU9.** utilities used in drug coating process
- KU10.** basic in-process checks for drug coating process
- KU11.** procedure of preparation of drug coating
- KU12.** procedure of drug coating

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to maintain proper and concise records as per given format and compliant with ALCOA principle
- GS3.** use verbal communication skills in communicating the details of work done to appropriate people and during interaction with regulatory inspectors and other teammates
- GS4.** use planning and organizing skills in every activity planned and performed in stent operations and to achieve resource optimization
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** use critical skills in analyzing impact of deviations, wastage and rejects to the environment and efficiency, compliance and cost
- GS8.** demonstrate decision making while making necessary adjustments in parameters to achieve quality specifications
- GS9.** apply customer-centricity while responding to auditors and QA personnel

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparation of Drug Coating Operations</i>	15	25	5	5
PC1. ensure availability of high quality raw material as per defined ISO Class	-	-	-	-
PC2. perform preparation of drug coating solution under reverse laminar air in dispensing booth of class 100 (ISO class 5) surrounded by class 10000 (ISO class 8)	-	-	-	-
PC3. ensure drug coating solution is prepared as per SOP	-	-	-	-
PC4. ensure cleanroom are periodically validated and periodic environmental monitoring is performed as per SOP	-	-	-	-
PC5. ensure drug coating solution is tested as per quality plan	-	-	-	-
<i>Drug Coating Operations</i>	15	25	5	5
PC6. ensure coating process is performed with a validated process in a controlled environment of class 10000	-	-	-	-
PC7. ensure pre cleaned stent identification number and its measured weight are recorded accurately	-	-	-	-
PC8. perform pouring of prepared solution into spray gun and start coating process	-	-	-	-
PC9. ensure vacuum drying process is performed in same controlled environment of class 10000	-	-	-	-
PC10. ensure cleanroom are periodically validated	-	-	-	-
PC11. ensure drug coating process is validated as per records	-	-	-	-
PC12. ensure sending the drug coated stent to next stage after quality inspection as per quality plan	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. ensure any rejected drug coated stents are modified if feasible else segregate and dispose them as per SOP	-	-	-	-
NOS Total	30	50	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6208
NOS Name	Perform Drug Coating Process
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	4.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023



Qualification Pack

LFS/N6209: Perform Crimping process

Description

The NOS is about performing crimping process

Scope

The scope covers the following :

- The NOS Scope is :
- Crimping Operations

Elements and Performance Criteria

Crimping Operations

To be competent, the user/individual on the job must be able to:

- PC1.** ensure availability of duly packed PTCA Balloon Catheters from raw material store
- PC2.** perform crimping process for stents in cleanroom under class 100 laminar air flow unit surrounded by clean room class 10000 environment as per SOP
- PC3.** take out balloon catheter from hoop tray or dispenser and remove protective sheath from it
- PC4.** mount coated stent on PTCA balloon catheter and position between radio opaque markers and observe with 10X magnifying lens
- PC5.** ensure that the measurement of final crimped stent is in compliance to the manufacturing plan
- PC6.** perform insertion of crimped stent system into hoop tray or dispenser
- PC7.** ensure approved stent system are sent to next stage for primary packaging after quality inspection as per quality plan
- PC8.** ensure segregation and disposal of rejected products as per SOP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** clean room behaviour practices and gowning procedures
- KU2.** SOP for entry and exit from manufacturing area
- KU3.** PPE used in crimping process and their work instructions
- KU4.** material segregation, handling and storage guidelines
- KU5.** type of machinery and equipment used for crimping process and their critical parameters
- KU6.** care and maintenance of equipment and machinery
- KU7.** removing and replacing components or parts of machinery and equipment
- KU8.** specialized techniques used in crimping of stent
- KU9.** utilities used in crimping process
- KU10.** basic in-process checks for crimping process



Qualification Pack

KU11. procedure of crimping ballon catheter with coated stent

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to maintain proper and concise records as per given format and compliant with ALCOA principle
- GS3.** use verbal communication skills in communicating the details of work done to appropriate people and during interaction with regulatory inspectors and other teammates
- GS4.** use planning and organizing skills in every activity planned and performed in stent operations and to achieve resource optimization
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** use critical skills in analyzing impact of deviations, wastage and rejects to the environment and efficiency, complianc and cost
- GS8.** demonstrate decision making while making necessary adjustments in parameters to achieve quality specifications
- GS9.** apply customer-centricity while responding to auditors and QA personnel

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Crimping Operations</i>	30	50	10	10
PC1. ensure availability of duly packed PTCA Balloon Catheters from raw material store	-	-	-	-
PC2. perform crimping process for stents in cleanroom under class 100 laminar air flow unit surrounded by clean room class 10000 environment as per SOP	-	-	-	-
PC3. take out balloon catheter from hoop tray or dispenser and remove protective sheath from it	-	-	-	-
PC4. mount coated stent on PTCA balloon catheter and position between radio opaque markers and observe with 10X magnifying lens	-	-	-	-
PC5. ensure that the measurement of final crimped stent is in compliance to the manufacturing plan	-	-	-	-
PC6. perform insertion of crimped stent system into hoop tray or dispenser	-	-	-	-
PC7. ensure approved stent system are sent to next stage for primary packaging after quality inspection as per quality plan	-	-	-	-
PC8. ensure segregation and disposal of rejected products as per SOP	-	-	-	-
NOS Total	30	50	10	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6209
NOS Name	Perform Crimping process
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	4.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

LFS/N6210: Perform Primary Packaging Process

Description

The NOS is about performing primary packaging process

Scope

The scope covers the following :

- The NOS Scope is :
- Primary Packaging Operations

Elements and Performance Criteria

Primary Packaging Operations

To be competent, the user/individual on the job must be able to:

- PC1.** perform sanitization and gowning procedures as per SOP and cleanroom guidelines
- PC2.** wear PPE before entering into the production area
- PC3.** ensure availability of QA approved containers, pouches and self adhesive labels
- PC4.** record the environmental variables like temperature and humidity
- PC5.** ensure inspection of final product before packaging
- PC6.** perform primary packaging of stent to avoid cross contamination
- PC7.** perform placing of stent in Tyvek Pouches as per SOP and regulatory guidelines
- PC8.** perform sealing of pouches using sealing machine at set temperature
- PC9.** ensure equipments are calibrated regularly
- PC10.** ensure inspection of sealed pouches for seal uniformity
- PC11.** ensure transfer of stent for further sterilization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** clean room behaviour practices and gowning procedures
- KU2.** SOP for entry and exit from manufacturing area
- KU3.** PPE used in primary packaging process and their work instructions
- KU4.** material segregation, handling and storage guidelines
- KU5.** type of machinery and equipment used for primary packaging process and their critical parameters
- KU6.** care and maintenance of equipment and machinery
- KU7.** removing and replacing components or parts of machinery and equipment
- KU8.** specialized techniques used in primary packaging of stent
- KU9.** utilities used in primary packaging process



Qualification Pack

- KU10.** basic in-process checks for primary packaging process
- KU11.** procedure of primary packaging
- KU12.** procedure of sealing of pouches

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use reading and comprehension skills to extract the relevant information from manuals, SOPs, health and safety instructions, memos, reports, job cards, and notes/comments
- GS2.** use written communication skills to maintain proper and concise records as per given format and compliant with ALCOA principle
- GS3.** use verbal communication skills in communicating the details of work done to appropriate people and during interaction with regulatory inspectors and other teammates
- GS4.** use planning and organizing skills in every activity planned and performed in stent operations and to achieve resource optimization
- GS5.** apply problem-solving skills to find solutions for workflow-related difficulties
- GS6.** apply critical skills to analyze and identify when to report an issue/concern to the supervisor and to deal with a colleague individually, depending on the type of concern
- GS7.** use critical skills in analyzing impact of deviations, wastage and rejects to the environment and efficiency, compliance and cost
- GS8.** demonstrate decision making while making necessary adjustments in parameters to achieve quality specifications
- GS9.** apply customer-centricity while responding to auditors and QA personnel

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Primary Packaging Operations</i>	30	50	10	10
PC1. perform sanitization and gowning procedures as per SOP and cleanroom guidelines	-	-	-	-
PC2. wear PPE before entering into the production area	-	-	-	-
PC3. ensure availability of QA approved containers, pouches and self adhesive labels	-	-	-	-
PC4. record the environmental variables like temperature and humidity	-	-	-	-
PC5. ensure inspection of final product before packaging	-	-	-	-
PC6. perform primary packaging of stent to avoid cross contamination	-	-	-	-
PC7. perform placing of stent in Tyvek Pouches as per SOP and regulatory guidelines	-	-	-	-
PC8. perform sealing of pouches using sealing machine at set temperature	-	-	-	-
PC9. ensure equipments are calibrated regularly	-	-	-	-
PC10. ensure inspection of sealed pouches for seal uniformity	-	-	-	-
PC11. ensure transfer of stent for further sterilization	-	-	-	-
NOS Total	30	50	10	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LFS/N6210
NOS Name	Perform Primary Packaging Process
Sector	Life Sciences
Sub-Sector	Medical Devices and In Vitro Diagnostic (IVD)
Occupation	Manufacturing - Medical Devices and IVD
NSQF Level	5
Credits	4.0
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	01/11/2026
NSQC Clearance Date	01/11/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by Life Sciences Sector Skill Development Council (LSSSDC)
2. Each Element will be assigned marks proportional to its importance in NOS. LSSSDC will also lay down the proportion of marks for Theory, Practical, Project, and Viva for each Element.
3. The assessment for the theory part will be based on the knowledge bank of questions created by the LSSSDC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. LSSSDC as assessment and awarding body will create unique evaluations for each assessment component i.e. theory, practical, project and viva for every student at each examination/training center based on this criterion.
6. Wherever any assessment component is not applicable/ feasible, the balance assessment components will be used to assess the candidate and accordingly the total marks will be calculated only for the applied



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assessment component.

7. To pass the Qualification Pack, every trainee should score a minimum of 50%-70% of marks in each NOS (depending on NSQF Level) to successfully clear the assessment. In the case of a Govt funded program, the program guidelines will be overarching on the pass percentage rules.

8. In case of unsuccessful completion, the trainee may seek re-assessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6203.Perform pre-production check and preparation of various machines for manufacturing	30	45	15	10	100	20
LFS/N0130.Ensure adherence to environment, health and safety guidelines in manufacturing facility and controlled cleanroom area	40	40	10	10	100	10
LFS/N0131.Ensure a hygienic and clean work area to avoid contamination in manufacturing facility	40	40	10	10	100	10
LFS/N6211.Perform reporting and documentation with data integrity for medical devices and IVD	40	40	10	10	100	10
LFS/N0133.Coordinate and communicate with supervisor/ engineer, teams and auditors	40	40	10	10	100	10

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	210	235	55	50	550	70

Elective: 1 Laser Cutting

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6204.Perform Laser Cutting Process	30	50	10	10	100	30
Total	30	50	10	10	100	30

Elective: 2 Descaling and Annealing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6205.Perform Descaling and Annealing process	30	50	10	10	100	30
Total	30	50	10	10	100	30

Elective: 3 Electropolishing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6206.Perform Electropolishing Process	30	50	10	10	100	30
Total	30	50	10	10	100	30

Elective: 4 Ultrasonic Cleaning

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6207.Perform Ultrasonic Cleaning Process	30	50	10	10	100	30
Total	30	50	10	10	100	30

Elective: 5 Drug Coating

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6208.Perform Drug Coating Process	30	50	10	10	100	30
Total	30	50	10	10	100	30

Elective: 6 Crimping

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6209.Perform Crimping process	30	50	10	10	100	30
Total	30	50	10	10	100	30

Elective: 7 Primary Packaging

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LFS/N6210.Perform Primary Packaging Process	30	50	10	10	100	30
Total	30	50	10	10	100	30



Qualification Pack

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
SOP	Standard Operating procedure
QA	Quality Assurance
QC	Quality Control
MSDS	Material Safety Data Sheet
SOP	Standard Operating Procedure
ISO	International Organization for Standardization
EHS	Environment, Health and Safety
SOP	Standard Operating Procedure
ISO	International Organization for Standardization
EHS	Environment, Health and Safety
QA	Quality Assurance
SOP	Standard Operating procedure
QA	Quality Assurance
QC	Quality Control
MSDS	Material Safety Data Sheet
SOP	Standard Operating Procedure
IVD	In-Vitro Diagnostic
PPE	Personal Protective Equipment
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate
QA	Quality Assurance
SOP	Standard Operating Procedure

Qualification Pack

ISO	International Organization for Standardization
EHS	Environment, Health and Safety
SOP	Standard Operating procedure
QA	Quality Assurance
QC	Quality Control
ISO	International Organization for Standardization
SOP	Standard Operating Procedure
ISO	International Organization for Standardization
EHS	Environment, Health and Safety
SOP	Standard Operating Procedure
ISO	International Organization for Standardization
EHS	Environment, Health and Safety
QA	Quality Assurance
SOP	Standard operating procedure
ISO	International Organization for Standardization
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate
QA	Quality Assurance
SOP	Standard Operating Procedure
QA	Quality Assurance
POSH	Prevention of Sexual Harassment
PwD	People With Disability
ALCOA	Attributable, Legible, Contemporaneous, Original and Accurate
EHS	Environment, Health and Safety
ISO	International Organization for Standardization
SOP	Standard Operating Procedure
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate

Qualification Pack

QA	Quality Assurance
ISO	International Organization for Standardization
SOP	Standard Operating Procedure
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate
QA	Quality Assurance
SOP	Standard Operating Procedure
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate
PPE	Personal Protective Equipment
SOP	Standard Operating Procedure
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate
QA	Quality Assurance
PPE	Personal Protective Equipment
SOP	Standard Operating Procedure
PPE	Personal Protective Equipment
QA	Quality Assurance
ISO	International Organization for Standardization
SOP	Standard Operating Procedure
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate
QA	Quality Assurance
PPE	Personal Protective Equipment
PTCA	Percutaneous transluminal coronary angioplasty
SOP	Standard Operating Procedure
PPE	Personal Protective Equipment
QA	Quality Assurance
SOP	Standard Operating Procedure
PPE	Personal Protective Equipment



Qualification Pack

QA	Quality Assurance
ALCOA	Attributable, Legible, Contemporaneous, Original, and Accurate

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
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